GMT

Green Metrics Technology

Buildable Areas for Laneway Homes

Midterm Status

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GMT

Green Metrics Technology Corp.

- Vancouver, BC
- Real estate decarbonization
- AI-powered building analytics

What They Do

Help housing managers, developers & modular manufacturers make data-driven decisions to optimize building performance, reduce carbon footprints, and simplify decarbonization investments.

BuildSense AI

AI-powered building management for multi-unit residential & commercial properties. Uses LLMs to create a digital twin, monitor performance in real-time, benchmark energy use, and trigger proactive maintenance before problems arise.

BuildBlox

Comprehensive platform for modular housing — covering Design (LCA from BIM/Revit), Manufacturing (real-time progress tracking), and Commissioning (proactive maintenance). Integrates operational data and financial insights to streamline the full build lifecycle.

Project Overview

"I want to build a laneway home in my property, but..."

1. Can I build the home?
2. What's the allowed area for me to build?
3. Are there other restrictions?

Why Is This Hard?

- BC municipalities publish zoning bylaws as dense PDF documents
- Rules are complex with nested conditions
- Each city uses a different format — no standard structure

Goal: Automate end-to-end — from raw PDF to applicable rules

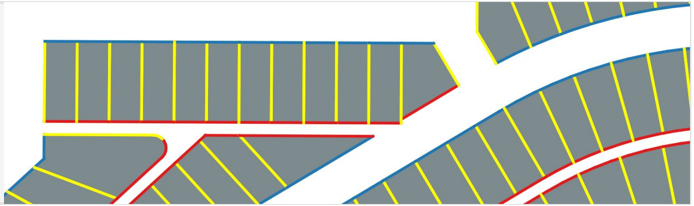
What Are We Trying to Solve?

Q1 **How to retrieve rule-relevant content from fragmented sources?**

→ *Self-developed rule extraction pipeline implementing LLM APIs*

Q2 **How to determine clashes between bylaws and building plans?**

→ *Using GIS — spatial overlay of setback rules onto parcel geometry*



Q3 **How to maintain end-to-end provenance linking outputs to the source?**

→ *Built-in: every rule cites an evidence_id linking back to its source PDF clause*

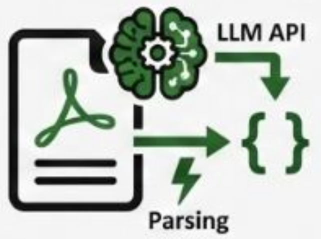
Scope: All cities across BC *Current progress: Finished Burnaby prototype*

System Architecture Overview

End-to-end pipeline: from raw PDF bylaw to buildable area output

Stage 1

Rule Extraction



PDF -> Structured JSON Rules via LLM

Stage 2

Verification Layer

Validating and Flagging extracted rules

Stage 3

GIS Rule Contract

Map setbacks, compute buildable area with provenance

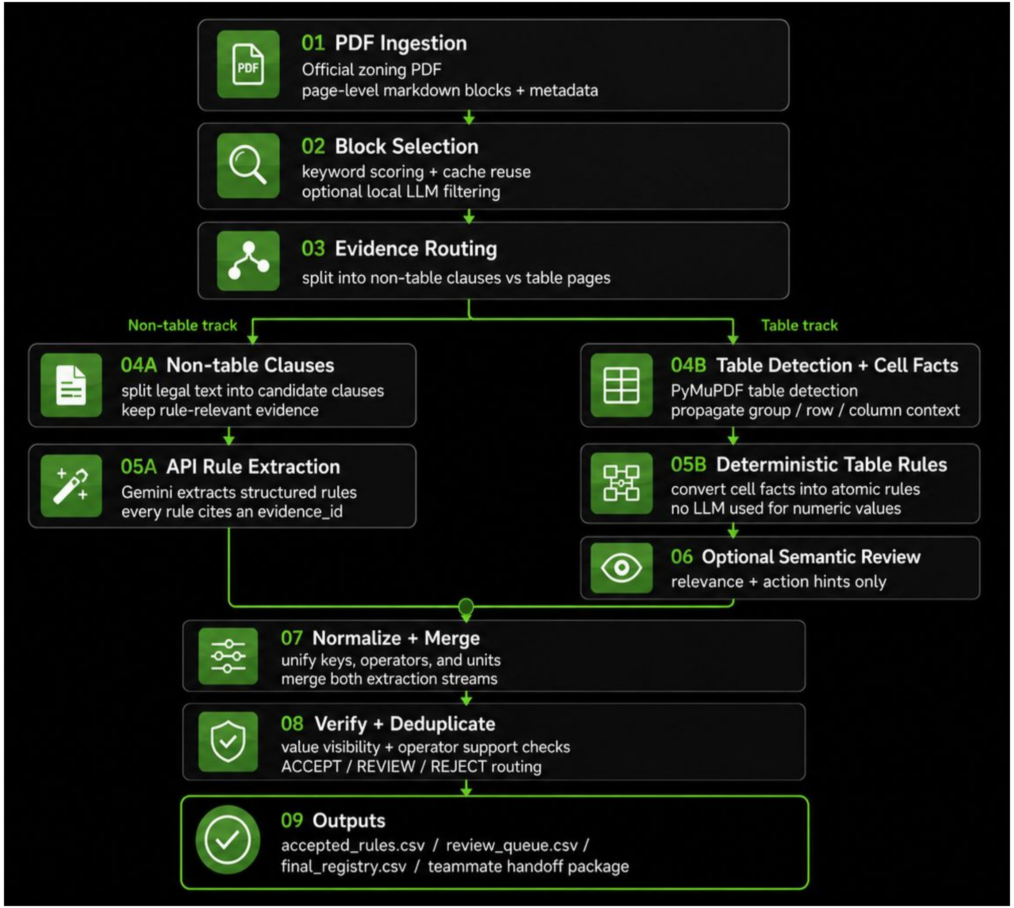
Output: Buildable area polygon per parcel • Setback distances applied • Linked back to source bylaw clause (provenance)

Stage 1: Rule Extraction

PDF Bylaw → Structured Rules via LLM Pipeline

Rule Extraction Pipeline — v.5

1. Convert zoning bylaw PDFs into structured rule data
2. Select rule-relevant blocks from official documents
3. Split evidence into two tracks:
   Text clauses ⟶ Gemini-based extraction
   Tables ⟶ deterministic PyMuPDF and VLM extraction
4. Normalize, merge, verify, and deduplicate rules
5. Output accepted rules and review queues



Example Output: Extracted Rule

Development Regulations

	Dwelling Type			
	Rowhouse	Small-Scale Multi-Unit		
Permitted Dwelling Units (including secondary suites)	1 to 3 Units	1 to 2 Units	3 to 4 Units	5 to 6 Units Frequent Transit Network Area Only
Minimum Lot Area	-	-	4 Units Only: 281 m^2	281 m^2
Maximum Lot Area [1]	280 m^2	-	-	-
Maximum Lot Coverage				
All Buildings	55%	Lots ≤ 567 m^2: 40% Lots > 567 m^2: 30%	40%	45%
Impervious Surfaces	70%	60%	60%	70%
Maximum Height				
Front Principal Buildings				
Height	sloping roof: 10 m | flat roof: 9.5 m			
Storeys (basement inclusive)	3 storeys	2.5 storeys	3 storeys	
Rear Principal Buildings				
Height	sloping roof: 7.5 m | flat roof: 7.0 m			
Storeys (basement inclusive)	2 storeys			
Accessory Buildings	4.0 m | 1 storey			
Minimum Lot Line Setbacks for All Buildings [2,3]				
Street Yard	3.0 m	front: 4.0 m | flanking: 3.0 m		
Lane Yard	1.5 m			
Interior Rear Yard	3.0 m, except 1.5 m for accessory buildings			
Interior Side Yard	0 m, except 1.2 m for end unit lots	1.2 m	1.2 m	1.2 m
Minimum Separation of Buildings on the Same Lot [4,5]				
Between Front Principals	-	2.4 m	2.4 m	2.4 m
Between Rear Principals	-	2.4 m	2.4 m	2.4 m
Between Front & Rear Principals	6.0 m			
Between All Other Buildings	2.4 m			

(B/L No. 14783-25-12-09)

Field	Value
normalized rule key	max_lot_coverage_all_buldings
rule object	All Buildings
operator	<=
value	30
unit	percent
condition	Small-Scale Multi-Unit / 1 to 2 Units / Lots > 567 m2
source stream	deterministic_ table
page number	2
verification_status	verified
final action	ACCEPT

Stage 2: Verification Layer

Validating Extracted Rules Against Source Documents

Stage 2: Verification Layer

Validating extracted rules against source documents

We treat all extracted rules as **untrusted candidates**.
This layer validates evidence and ensures **only trustworthy rules** move forward.

Input

Extracted rule candidates with evidence

→

Verification (Validates Evidence)

Deterministic Checks
Validate values, units, operators, scope, applies_to, etc.

Table-Aware Validation
Confirm table title, row/column headers, and cell values support the rule.

Conceptual Consistency
Ensure rule logic, scope, and applicability are consistent and non-contradictory.

Uncertainty Detection
Identify ambiguity, missing evidence, or conflicts that require human review.

→

Outcomes

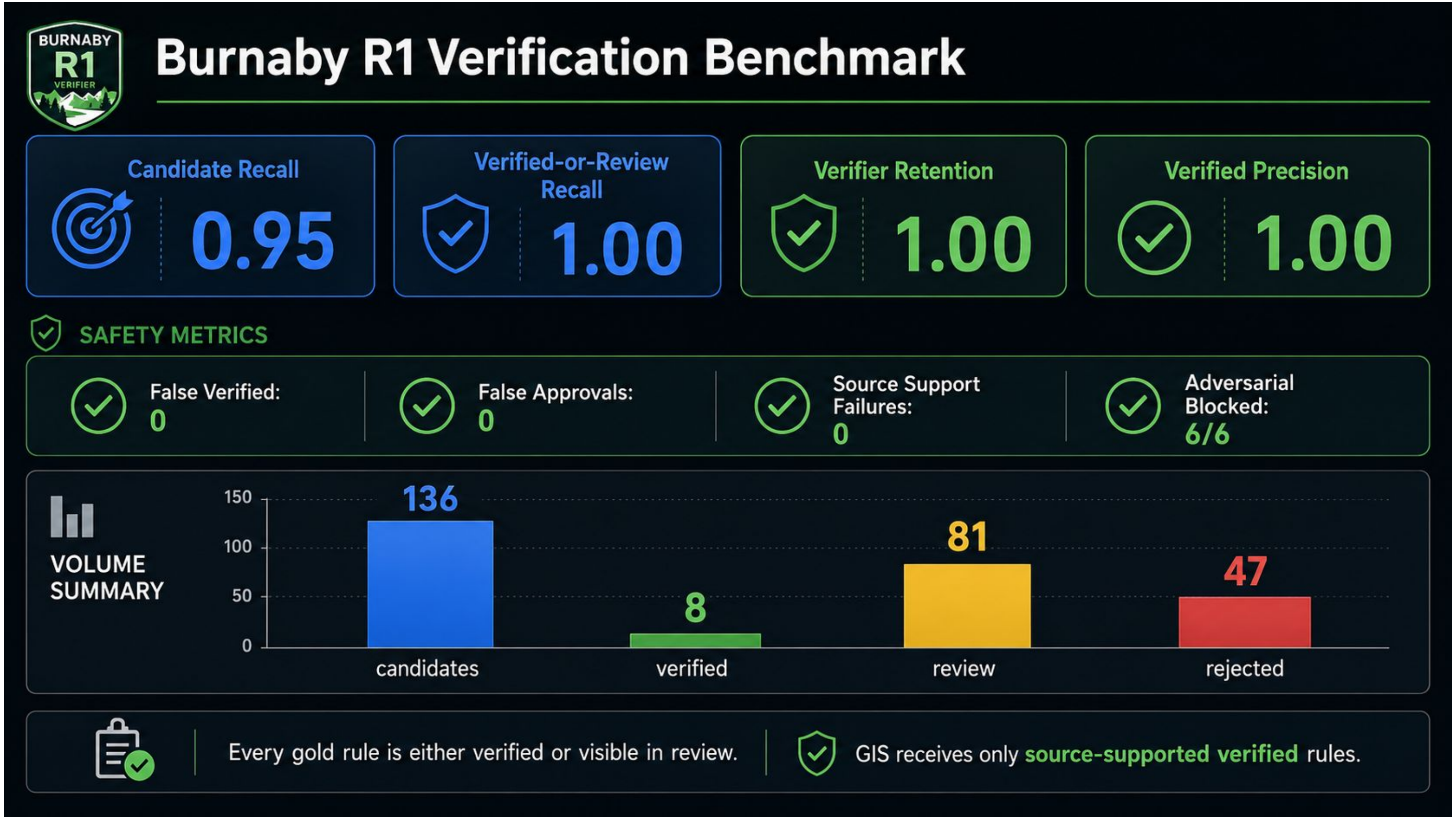
Verified
High-confidence rule proceeds downstream.

Review Needed
Uncertain or ambiguous → routed to human review.

Rejected
Not supported or contradicted by source.

Every decision includes an evidence link
back to the exact source PDF clause for full explainability and traceability.

Validation Results and Controls



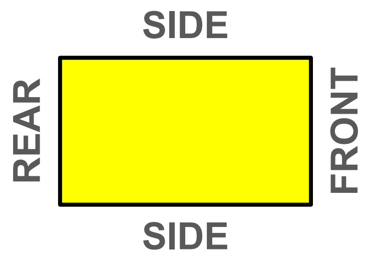
Stage 3: GIS Rule Contract

Buildable Area

Given a set of setback distance rules, how much space is left to build on?

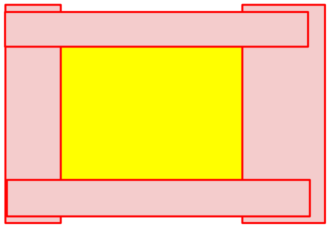
1 Label Boundaries

→ Label each boundary line as FRONT, SIDE, or REAR



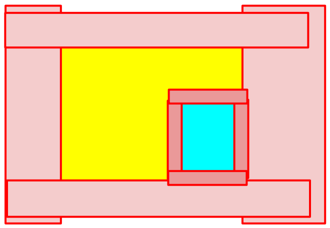
2 Apply Setbacks

→ Apply setbacks based on labels to get empty-lot buildable areas



3 Subtract Structures

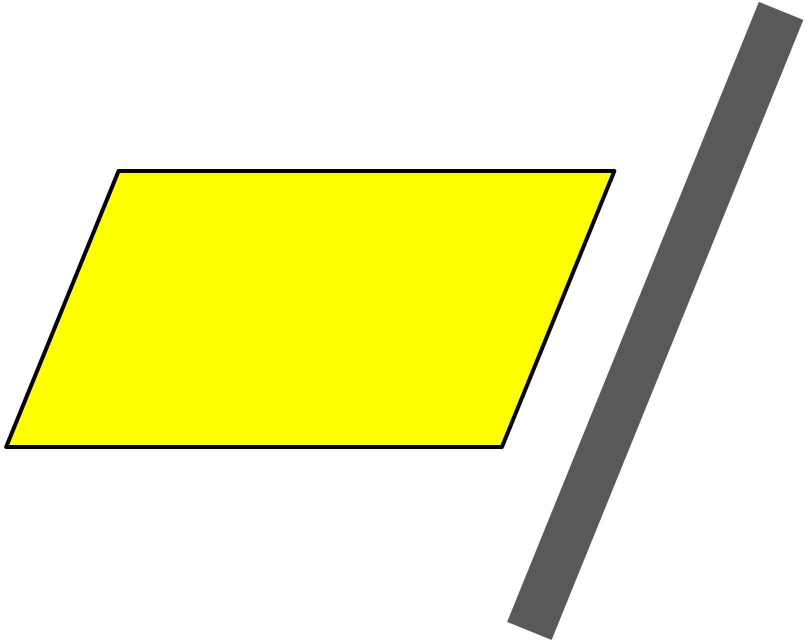
→ Subtract areas based on existing structures to get true buildable areas



Labelling Lot Boundaries

- Labelling is simpler for 4-sided parcels, but this only accounts for 79% of lots

- Labelling can be handled generically by considering which direction each lot boundary faces



Challenges During the Project

Cross-City Rule Standardization

Burnaby, Surrey, and Vancouver each use different terminology and implicit conditions. Building a schema that works across all cities without city-specific hard coding required significant iteration

PDF Table Parsing Complexity

Every city formats bylaw tables differently. Merged cells and multi-row labels caused rules to be duplicated or silently dropped

LLM Reliability on Legal Text

Local models struggled with complex legal tables. After switching to Gemini, verified accuracy — suggesting that using a smarter, more capable model could push accuracy higher

GIS Algorithm

It is difficult to determine which setback rules to apply for irregularly shaped lots

Next Steps

Successfully extracted required rules for Burnaby — now expanding to other cities

Next Goal

- Develop a more generalized and reliable extraction pipeline with improved accuracy, consistency, and scalability
- Use extracted rules with labelled lot boundaries to defined buildable areas